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UUCG RNA Tetraloop as a Formidable Force-Field Challenge for MD Simulations

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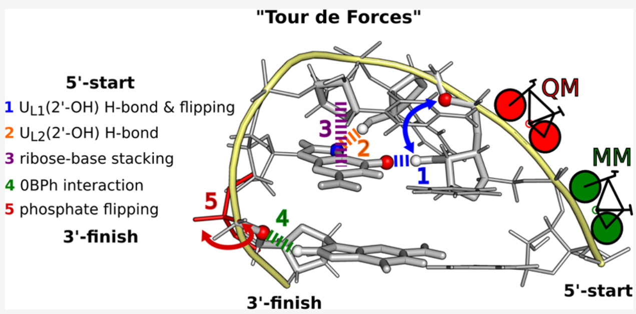
ABSTRACT: Explicit solvent atomistic molecular dynamics (MD) simulations represent an established technique to study structural dynamics of RNA molecules and an important complement for diverse experimental methods. However, performance of molecular mechanical (MM) force fields (ffs) remains far from satisfactory even after decades of development, as apparent from a problematic structural description of some important RNA motifs. Actually, some of the smallest RNA molecules belong to the most challenging systems for MD simulations and, among them, the UUCG tetraloop is saliently difficult. We report a detailed analysis of UUCG MD simulations, depicting the sequence of events leading to the loss of the UUCG native state during MD simulations. The total amount of MD simulation data analyzed in this work is close to 1.3 ms. We identify molecular interactions, backbone conformations, and substates that are involved in the process. Then, we unravel specific ff deficiencies using diverse quantum mechanical/molecular mechanical (QM/MM) and QM calculations. Comparison between the MM and QM methods shows discrepancies in the description of the 5′-flanking phosphate moiety and both signature sugar−base interactions. Our work indicates that poor behavior of the UUCG tetraloop in simulations is a complex issue that cannot be attributed to one dominant and straightforwardly correctable factor. Instead, there is a concerted effect of multiple ff inaccuracies that are coupled and amplifying each other. We attempted to improve the simulation behavior by some carefully tailored interventions, but the results were still far from satisfactory, underlying the difficulties in development of accurate nucleic acid ff's.



■ INTRODUCTION

Explicit solvent atomistic molecular dynamics (MD) simulations represent an established tool to study structural dynamics of RNA molecules.[1−3] MD simulations can complement diverse experimental methods by, e.g., allowing to go beyond the typical static (ensemble-averaged) description of RNA molecules obtained by structural biology techniques. However, reliability of MD predictions depends on quality of the used molecular mechanical (MM) force fields (ff's). Unfortunately, performance of these ff's remains far from satisfactory despite ongoing development and is also very system-dependent.[4,5] Many RNA molecules and protein/RNA complexes are quite well described by MD simulations, at least when starting the simulations from experimental structures. However, other systems progressively deteriorate upon extending the simulations. Surprisingly, even small RNA molecules may present persisting challenges for MD simulations, and among them, the UUCG tetraloop (TL) is saliently difficult.[6−11]

RNA tetraloops (TLs) are stem−loop RNA structures, which are common secondary structure elements in RNA. TLs participate in tertiary folding, RNA−RNA and protein−RNA interactions, ligand binding, and other important biological functions.[12,13] The UNCG TL (N stands for any nucleotide) is one of the two most abundant TLs.[14−16] It is thermodynamically very stable and possesses a clearly defined dominant folded topology stabilized by a set of signature molecular interactions and other structural features (Figure 1); by signature interactions, we mainly mean H-bonds that determine the sequence signature of the RNA motifs.[17] The native state of UNCG TL remains difficult to be described by the current RNA ff's. Recent folding simulations of UUCG TL indicated a large free-energy imbalance between native (folded) and other (misfolded/unfolded conformations) states.[6−10,18−20] In addition, the characteristic UUCG native structure is lost in sufficiently long standard simulations. Two different effects were deemed to dominantly lead to the incorrect folded/unfolded free-energy balance, namely, (i) excessive stabilization of the unfolded ssRNA structure by

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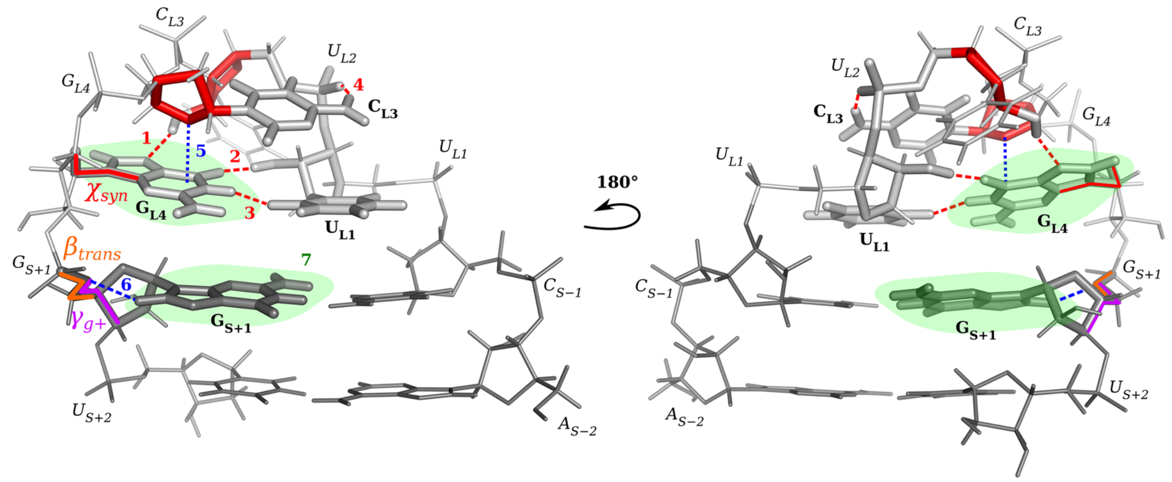

Figure 1. Native structure of the UUCG TL. Nucleotides of the loop are labeled as U_{L1}, U_{L2}, C_{L3}, and G_{L4}, and stem nucleotides are labeled as A_{S-2}, C_{S-1}, G_{S+1}, and U_{S+2}. Phosphates and bases are identified by normal and bold font, respectively. r(acUUCGgu) and r(gcUUCGgc) 8-mers were studied in this work. Key structural features of UUCG TL are colored in red and include the U_{L2}(2′-OH)···G_{L4}(N7) H-bond (marked as "1" in the figure), U_{L1}(2′-OH)···G_{L4}(O6) H-bond (2), U_{L1}(O2)···G_{L4}(N1) H-bond (3), C_{L3}(N4H)···U_{L1}(*pro*-R_P) H-bond (7BPh interaction[37]) (4), syn conformation of G_{L4} nucleotide (χ_{syn}), and C2′-*endo* pucker of U_{L2} and C_{L3} ribose. Other key structural features that were investigated in this work include the C_{L3}(O4′)···G_{L4}(base) perpendicular ribose−base stacking (5), G_{S+1}(C8H8)···G_{S+1}(O5′) 0BPh interaction (6), mutual position of G_{L4} and G_{S+1} nucleobases (7), and β_{trans} (orange) and γ_{g+} (violet) conformations of the G_{S+1} nucleotide.

intramolecular base−phosphate and sugar−phosphate interactions and (ii) destabilization of the native folded state by underestimation of the native H-bonds including the stem base pairing.[8,9,21,22] Recently, we introduced a general external potential tuning H-bond interactions (gHBfix),[8,19] which is used as an additional ff term to improve RNA simulations. Even with this modification, description of UUCG TL remained imbalanced and we were not capable to fold the r(gcUUCGgc) 8-mer TL.

In the past, there have been many attempts to simulate the UNCG TL (reviewed extensively in ref 4). Retrospectively, earlier studies reporting good performance of the simulations did not identify the true extent of the ff imbalances. Probably, the first successful folding of UUCG TL was reported by Chen and Garcia.[18] They have used an AMBER ff with modified van der Waals (vdW) parameters (namely, rescaling of base−base and base−water interactions) and performed temperature replica-exchange MD (T-REMD) simulations with 64 replicas starting from unfolded states of the r(gcUUCGgc) 8-mer TL. Several different folding pathways were observed, and the folded cluster with all signature interactions formed had a centroid of around 0.8 Å root-mean-square deviation (RMSD) from the native UUCG TL structure. Authors reported population of the folded state slightly above 20% at 300 K.[18] However, it was subsequently shown[6,23] that the 4 Å RMSD criteria used for monitoring the overall population of the folded state may have overestimated the folded fraction to an unknown extent. Bergonzo et al. used this Chen−Garcia ff in massive multidimensional REMD (M-REMD) folding simulations of r(cacUUCGgug) TL (with 360 replicas), where the two terminal base pairs of the stem were stabilized by Watson−Crick H-bonding restraints to improve the convergence. They have reported ∼10% population of properly folded TL at 277 K and less than 1% at 300 K.[9] Thus, the extent to which the Chen−Garcia ff can fold UNCG TLs is still not fully clear. More importantly, it was shown that the Chen−Garcia ff introduced overly strong H-bonding between bases and some other undesired side effects for other systems[9,24] and thus the ff is not widely used in simulations of common RNA systems.[4] Reparameterization of nine dihedral terms introduced by Aytenfisu et al.[11] did not provide any improvement in the description of the UUCG TL. The authors actually reported in the original paper that their ff could even speed up disruption of the r(ggcacUUCGgugcc) 14-mer (PDB ID 2KOC[25]) in comparison with the standard AMBER RNA ff99bsc0χ_{OL3}[26−29] ff during 2.5-μs long standard MD simulations.[11] Modifications of the Lennard−Jones combining rules via nonbonded fix (NBfix) to balance the RNA−RNA interaction by Yang et al.[30] resulted in some moderate improvement in folding simulations of r(ccUUCGgg) 8-mer, i.e., the folding/unfolding free-energy imbalance was somewhat reduced. However, authors explicitly stated that their nonbonded term tuning is still not sufficient to correctly fold the 8-mer TL.[30] In addition, it was later shown that this ff has problems with the proper description of key RNA tertiary interactions like the A-minor interactions. When used to simulate RNA Kink-turn Kt-7, the simulations resulted in a complete loss of its characteristic bent shape and then complete structural degradation.[19]

Some success has been recently achieved with the DESRES ff for the r(ggcacUUCGgugcc) 14-mer with a longer A-RNA stem in combination with CHARMM22 ion parameters.[31] Extended simulated tempering folding simulations suggested a fraction of the folded native state around 30% at 300 K. We have later carried out standard simulations of the r(ggcacUUCGgugcc) 14-mer with the DESRES ff, which indicate a population of the folded UUCG TL of around 30−40% with CHARMM22 ions.[32] These simulations were initiated from the folded state, extended to 20 μs, and, despite being not fully converged, documented several unfolding−refolding events. The A-RNA stem was entirely stable on this time scale. Surprisingly, when using Joung−Cheatham ions, the native TL structure appeared to be irreversibly lost, which is a very rare case of documented sensitivity of RNA simulations to details of monovalent ion parameters.[32] Further, the DESRES ff is not able to fold the shorter gcUUCGgc 8-mer sequence.[19,32] However, the main problem is that the DESRES ff completely disrupts (including secondary structure rearrangements) some other important RNA systems even in standard simulations on the sub-microsecond time scale, irrespective of the ions used.[19,32] One should always keep in mind that side effects in simulations of functional RNA molecules, which may result from overtraining to model systems, outweigh partial improvements for small models.[4,19,32] As another attempt, Cesari et al. suggested a methodology for refinement of existing RNA ff's using simulations and primary NMR data for small model systems.[20] They demonstrated that modulating dihedral terms of the AMBER ff by small dihedral corrections (~1 kcal/mol) somewhat reduces the free-energy difference between the folded and unfolded states of r(ccUUCGgg) 8-mer TL. The free-energy imbalance was actually reduced even compared to simulations with the DESRES ff. Nevertheless, the adjustment was not sufficient to stabilize the 8-mer UUCG folded state.[20] In summary, we are not aware of any existing RNA ff able to properly describe the UUCG TL with all its signature interactions that could be considered as a ff generally applicable to folded RNA molecules. The widely tested AMBER ff99bsc0χ_{OL3} ff,[29] albeit not perfect, still represents a valid choice for simulations of common RNA molecules and protein−RNA complexes and thus is used as the starting point in our work.

In the present work, we analyze origins of the notoriously problematic description of the UNCG TL. All simulations were done with the UUCG sequence, but results reported here should be applicable to all UNCG tetraloops. First, we carry out a detailed analysis of our earlier and new MD simulations and obtain a specific picture of the sequence of events that leads to loss of the native state. We identify molecular interactions, backbone substates, and other conformational features that are involved in the process. Then, we try to unravel specific ff deficiencies using massive quantum mechanical/molecular mechanical (QM/MM)[33] and QM calculations. Obviously, as discussed elsewhere,[34−36] QM and QM/MM calculations cannot replace MD simulations due to the lack of Boltzmann sampling. There is generally a limited transferability between calculations of potential energy surfaces (gradient optimizations or energy scans) as done by QM methods and free energies that determine populations in MD simulations. Still, careful comparison of accurate-enough QM and QM/MM calculations with equivalent ff data provides substantial insights into the imbalances that are inherent to ff's. This is the goal of our QM/MM and QM computations. Finally, we try to improve the simulation behavior by some carefully tailored interventions, with a partial success. Our work clearly indicates that poor behavior of the UUCG simulations is a complex issue and cannot be explained by one single dominant factor that would be straightforwardly correctable. Instead, there is a concerted effect of multiple ff inaccuracies that are coupled and magnifying each other's effect. Such accumulation of diverse inaccuracies in description of a short UUCG RNA sequence that in addition folds into a very stiff structure provides no conformational freedom to buffer the individual imbalances. It leads, even for moderate imbalances, to substantial structural strains.

■ METHODS

Starting Structures. The initial coordinates were taken either from a high-resolution NMR structure of the r(ggcacUUCGgugcc) 14-mer (PDB ID 2KOC[25]) or the r(gcUUCGgc) 8-mer was excised from a 2.8 Å resolution X-ray structure (PDB ID 1F7Y[38]). The starting topologies and coordinates for MD simulations were prepared using the tLEaP module of the AMBER 16 program package.[39] The structure of the isolated small TL motif in solution is in dynamic temperature-dependent equilibrium between folded and unfolded conformations. Most of the calculations were done using 8-mer oligonucleotides, resulting, upon correct folding, into a two-base-pair stem and the TL itself. As TLs with longer stems require higher temperatures for melting (unfolding),[40,41] minimal eight-nucleotide long (8-mers) TL motifs with just two base pairs are commonly used in computational studies.[6−10,18,19] There is a clear experimental evidence that even these short oligomers should be dominantly in the folded state,[42−44] validating the 8-mer calculations.

Classical MD Simulations. We used the standard ff99bsc0χ_{OL3}[26−29] AMBER RNA ff with the vdW modification of phosphate oxygens developed by Steinbrecher et al.,[45] where the affected dihedrals were adjusted as described elsewhere.[22] This RNA ff is abbreviated as χ_{OL3CP} henceforth, and libraries can be found in the Supporting Information of ref 8. In the simulations, we additionally applied our gHBfix[19] H-bonding potential that was shown to improve the overall performance of the RNA ff. In most cases, we used its basic version from ref 19, abbreviated subsequently as gHBfix19 in ref 46. When any modified alternative is used, it is always specified in the text.

The OPC[47] water model and the Joung−Cheatham[48] ion parameters (~0.15 M KCl) were used in majority of simulations. The choice of the OPC water model as the main water model for our study can be justified in the following way. It is more sophisticated than the TIP3P or SPC/E water models, and it is therefore expected that when improving the solute ff, this water model can be better balanced with it. Further, it has been shown that the OPC water model improves simulations of some RNA systems. More specifically, it significantly improves the outcome of r(GACC) and r(CCCC) tetranucleotide simulations using AMBER ff99bsc0χ_{OL3} ff with the modified phosphate oxygens (similar to the present χ_{OL3CP} but without the phosphate dihedral refit).[49] In addition, our gHBfix potential was developed directly for the OPC water. When we tried to find suitable gHBfix parameters with the less sophisticated water models (TIP3P and SPC/E), we found considerable difficulties in tuning the gHBfix term; see our previous work[46] for some discussion of performance of other water models in combination with the gHBfix potential. However, we do not suggest that the OPC water is the ultimate choice for MD simulations of nucleic acids as, e.g., the OPC water model is rather suboptimal for G-quadruplex simulations, being inferior to SPC/E.[50] The outcome of simulations always depends on mutual compensation of errors of different ff terms. Generally, the OPC water model appears to weaken some intramolecular interactions. This may be profitable mainly in folding simulations as it may prevent the simulated molecules to be trapped in misfolded or overcompacted structures. However, it can be also accompanied with undesired destabilization of some native interactions. Thus, specific tests in this paper involved TIP3P[51] and SPC/E[52] water models and were also performed in Na^+ net-neutral and 1.0 M KCl salt-excess ion environments. Other details about the MD protocols can be found in Section S1 in the Supporting Information. Where possible, we used also simulations from our preceding papers;[19,32] see Table S1 in the Supporting Information for a

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complete list of standard MD simulations analyzed in this work.

REST2 Settings. The replica-exchange solute tempering (REST2)[53] simulations of the r(gcUUCGgc) TL were performed at 298 K (the reference replica) with either 12 or 16 replicas. Details about settings can be found elsewhere.[8] The scaling factor (λ) values ranged from 1 to 0.59984 and from 1.0454 to 0.59984 for 12 and 16 replicas, respectively. These values were chosen to maintain an exchange rate above 20%. The effective solute temperature ranged either from 298 K (12 replicas) or 285 K (16 replicas) to ~500 K. The hydrogen mass repartitioning[54] with a 4 fs integration time step was used. See Table S2 in the Supporting Information for list of enhanced sampling simulations.

QM/MM, QM/COSMO, MM, and MM/GB Geometry Optimizations of UUCG TL Conformations. We carried out a series of gradient geometry optimizations to compare the MM (i.e., ff) description with advanced QM methods. The initial r(acUUCGgu) and r(gcUUCGgc) 8-mers were either taken from eight different MD snapshots (structures **1**−**8**; see the Supporting Information for structures) or excised from the NMR structure of the r(ggcacUUCGgugcc) 14-mer (PDB ID 2KOC, structure **9**). The counterions (K^+) were added using the tLEaP module of AMBER 16 to neutralize the RNA. An SPC/E water model sphere with radius ~40 Å from the geometrical center of the RNA was prepared using tLEaP for all of the initial structures. Short solvent equilibrations (<10 ps MD) were performed for all of the structures **1**−**9**. In addition, five different water distributions were used for geometry optimizations of structures **5**, **6**, **7**, **8**, and **9** obtained by 20, 40, 60, 80, and 100 ps long water equilibrations. The resulting 25 starting structures are denoted **5a**−**e**, **6a**−**e**, **7a**−**e**, **8a**−**e**, and **9a**−**e**. Water distribution may influence the relaxed solute structures, as shown in ref 34. For the sake of simplicity, only results for structures after the 100 ps long water equilibrations (i.e., structures **5e** to **9e**) are documented in the main text, while the full set of data can be found in the Supporting Information.

The additive QM/MM scheme with point-charge approximation for electrostatic embedding was used for all QM/MM calculations. The QM/MM module of AMBER 14[55,56] was coupled with TURBOMOLE V7.3[57,58] using an in-house modified version of Sander[34,59,60] from the AMBER 14 program package. For geometry optimization, the limited-memory Broyden−Fletcher−Goldfarb−Shanno (L-BFGS) quasi-Newton algorithm[61] was used with a convergence threshold of 1×10^{-4} kcal/(mol·Å) for the gradient norm. Turbomole was used to calculate energies and gradients of the QM region using the composite PBEh-3c[62] hybrid-DFT method with resolution-of-identity (RI) approximation for the Coulomb integrals.[63] The PBEh-3c method employs the double-ζ valence-polarized def2-mSVP basis set, the empirical DFT-D3 dispersion correction[64] with Becke−Johnson (BJ) damping,[65] and the geometrical counterpoise (gCP) correction for intra- and intermolecular basis set superposition errors (BSSEs).[66] The QM region included the whole RNA 8-mer, while the MM region consisted of water molecules and counterions.

Similarly to the work by Pokorná et al.,[34] QM/MM optimizations were mirrored by equivalent MM optimizations from identical starting structures. For the sake of completeness, we note that the ff99bsc0χ_{OL3} was used for structures **1**−**5** and **9** and χ_{OL3CP} for structures **6**−**8**; the ff difference is, however, marginal for optimizations (see Figure S1 in the Supporting Information). Geometrical parameters (backbone dihedrals, intramolecular distances, and signature H-bonds) of the QM/MM- and MM-optimized TLs were analyzed using the CPPTRAJ[67] module of AMBER 16. Mutual orientation of nucleobase planes, i.e., the angle between planes of the G_{L4} syn base and the closing G_{S+1} base, and the sugar−base stacking interaction were analyzed using the Geom_util program (https://github.com/hokru/geom_util).

Besides MM optimizations from the initial structures, we also performed MM reoptimizations starting from the QM/MM-optimized structures to reduce potential uncertainty due to comparison of too distinct local minima, i.e., a situation where the QM/MM and MM optimizations from the MD snapshot diverge to nonequivalent (too different) local minima.[34] The QM/MM to MM reoptimized geometry should typically stay close to the QM/MM-optimized geometry (see the Supporting Information). Therefore, comparing geometrical parameters between QM/MM-optimized and QM/MM to MM reoptimized geometries should conservatively show differences between QM and MM potential energy surfaces (PESs), although the reoptimization may also mask MM imbalances by excessively attenuating the rearrangements; for detailed discussion, see ref 34.

RNA 8-mers were additionally optimized in implicit solvents. The conductor-like screening model (COSMO)[68] and the generalized Born (GB) model[69,70] were used for QM and MM, respectively, with a dielectric constant ε of 78.5.

Relaxed U_{L1}(2′-OH) Dihedral Scans of RNA TLs. To analyze quality of the MM description of orientation of the key U_{L1}(2′-OH) group, we compared QM and MM relaxed conformational energy scans of the U_{L1}(C1′−C2′−O2′−O2′H) dihedral for 8-mer structures **2**, **7**, **8**, and **9**.

First, the ONIOM scheme[71] was used for MM optimizations of the UUCG TL 8-mer (the high-layer) and solvent (the low-layer) using Gaussian 09.[72] As the same MM method (χ_{OL3CP} RNA ff) was used for both layers, the energy/gradient components of the high-layer calculations effectively cancel out. This approach results in MM geometry optimization with constraints using the microiterations approximation where water molecules are relaxed after each dihedral angle change. This is a technical workaround to enable efficient internal coordinate quasi-Newton optimizations necessary for constraints for an otherwise prohibitively large molecule. We mark this approach as MM/MM optimization throughout the paper.

To save computation time, the MM/MM-optimized geometries were used to initiate the corresponding QM/MM optimizations. The AMBER 14 coupled to Turbomole was used for QM/MM optimizations by applying tight geometry restraints on the U_{L1}(C1′−C2′−O2′−O2′H) dihedral (see the Supporting Information for details). Technical difference between using geometry constraints and restraints has no practical impact on the results.

QM and MM Calculations of Small Model Systems. We also carried out calculations on various small model systems. We extracted specific structural moieties of the UUCG TL (see Section S2 in the Supporting Information for the list of studied small models) for comparative QM and MM analyses, i.e., local geometry optimizations, interaction energy scans, and conformational energy scans. The Xopt[73,74] tool coupled with Turbomole was used for all of the QM geometry optimizations performed at the PBEh-3c level using the RI approximation. Turbomole was also used for the rigid-

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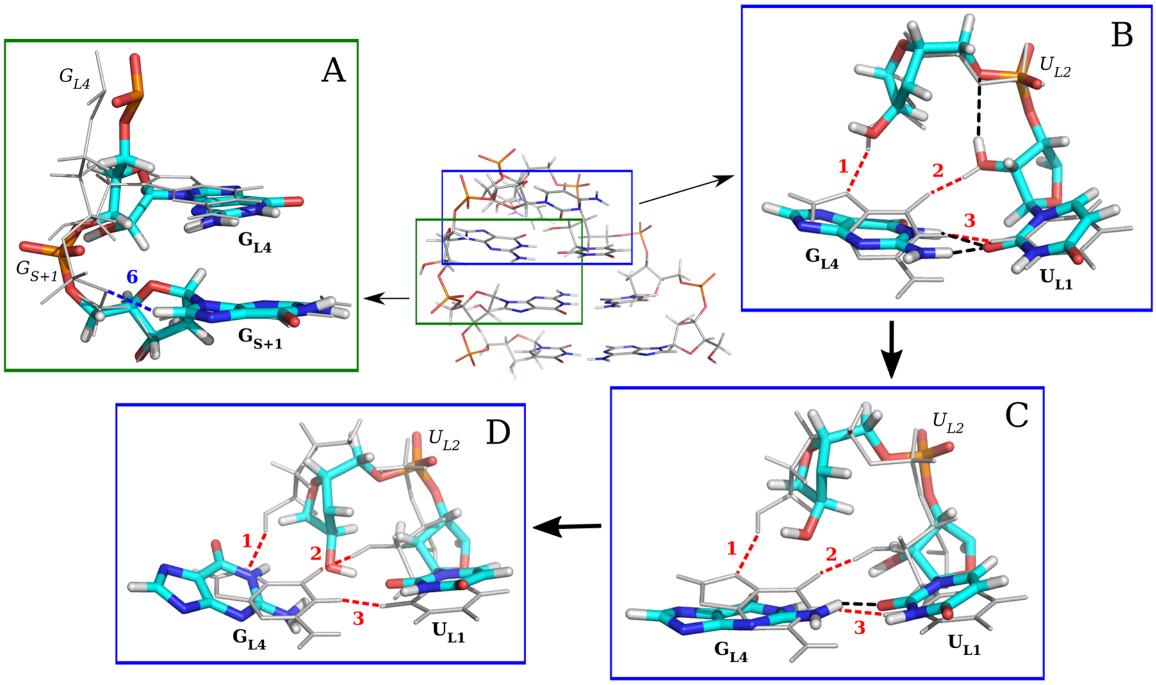

Figure 2. Disruption of the UUCG TL in MD simulations caused by imperfect description of several intramolecular interactions within AMBER ff. The UUCG TL is labeled similarly to Figure 1 except that intermediate H-bonds are depicted by black dashed lines. The native structure of the 8-mer r(acUUCGgu) TL is shown in the middle with two important regions: (i) G_{L4}−G_{S+1} dinucleotide moiety (green) and (ii) tetranucleotide loop (blue). Insets of (A)−(D) show only the specific fragments, where details about loss of important interactions or key conformational characteristics are highlighted. The overlaid native structure is depicted by gray lines in each of the insets. The important interactions 1−3 and 6 of the native UUCG TL are marked in a similar way to that in Figure 1 for each of the insets. The disruption pathway of the UUCG TL is depicted by insets (B)−(D), and its flow is indicated by arrows. All of the interactions and nucleotides shown by the insets are labeled similarly to Figure 1. (A) G_{L4}−G_{S+1} dinucleotide with a flipped G_{S+1} phosphate accompanied by loss of the 0BPh interaction (6). (B) Part of the loop region, where the U_{L1}(2′-OH) group is oriented toward the U_{L2} phosphate forming the spurious U_{L1}(2′-OH)···U_{L2}(O5′) H-bond. The partial twist of the G_{L4} nucleobase leads to the formation of the bifurcated G_{L4}(N1H/N2H)···U_{L1}(O2) H-bond. Signature H-bonds 1 and 2 (see Figure 1) are lost. (C) Signature H-bonds (1−3) are lost and G_{L4} is forming only the spurious G_{L4}(N2H)···U_{L1}(O2) H-bond (black dashed line). (D) G_{L4} is further repelled from the loop and is not stabilized by any H-bond (bulge-out state). The common disruption pathway seen in simulation is documented as a movie in the Supporting Information.

monomer QM energy scans performed at the same level of theory. ORCA V4.1.0[75,76] was used for the reference (highest-quality) QM calculations performed at the DLPNO-CCSD(T)/TightPNO/CBS level using aug-cc-pV(T-Q)Z and corresponding auxiliary aug-cc-pVQZ/C basis sets and with TightSCF setting.[77−81] MM geometry optimizations were performed using Xopt coupled with AMBER 14 or using Gaussian 09. QM and MM optimizations and single-point energy calculations of dinucleoside monophosphate models were performed with COSMO and GB implicit solvents (ε = 78.5). For rigid-monomer MM interaction energy scans, we used the development version of the bff program[82] using nonbonded energy terms of the AMBER ff. For the energy scans of small models, the MM method is denoted AMBER ff, and details are specified in the Supporting Information. Gaussian 09 was used for the MM relaxed conformational energy scan of the O3′-methylated-guanosine-monophosphate model. For summary of all QM and QM/MM calculations and additional computational details on calculations of the small models, see Sections S3 and S4 in the Supporting Information.

We used programs Molden,[83] VMD,[84] and PyMOL[85] for visualizing and analyzing the structures.

■ RESULTS AND DISCUSSION

We applied diverse computational techniques, namely, standard MD simulations, enhanced sampling REST2 simulations, QM, MM, and multiscale QM/MM calculations (see Methods) to understand limitations of current state-of-the-art AMBER RNA ff's in description of the UNCG TL. First, we analyzed disruption (unfolding) paths from standard as well as enhanced sampling MD simulations. We identified common features that seem to be responsible for instability of the native topology, i.e., specific structural reconformations that precede the collapse of the loop. We then performed various QM/MM and MM calculations of the UUCG TL to analyze differences in backbone dihedrals, signature H-bonds and other structural features, as described by QM and MM to assess potential ff problems. Furthermore, we applied QM and MM calculations to small model systems to study interactions that may be triggering structural changes within the loop. Finally, we proposed certain fixes, namely, (i) modulation of the key 0BPh interaction, (ii) adjustment of the sugar−base signature H-bonds, and (iii) reduction of the vdW radii of nonpolar hydrogens, which were subsequently tested during series of enhanced sampling MD simulations. In the text, we first describe in detail the loss of the TL's native structure in MD simulations and identify the individual potentially problematic energy terms. Then, we present the specific investigations visualizing and quantifying the ff limitations with aid of QM methods. Finally, we describe modifications of the ff's and their tests in simulations. Although we use only UUCG TL for all analyses, our results should be applicable to all UNCG TLs.

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Anatomy of Disruption of the UUCG TL in Simulations: A Progressive and Undesired Loss of Several Interactions. We inspected specific parts of trajectories from standard as well as enhanced sampling MD simulations of the UUCG TL from our previous work,[19] where signature interactions within the UUCG TL are lost and G_{L4} is repelled from its native conformation (Figure 2). Interestingly, disruption paths using multiple versions of the AMBER RNA ff[19,26,28,29] have common features and could be characterized as an interplay of several factors. These factors involve interactions that are, apparently, wrongly described with current RNA ffs.

First important factor is conformation of the G_{S+1} phosphate that is forming (in its native state) a 0BPh interaction[37] (Figure 2A). The sugar–phosphate backbone around the G_{S+1} phosphate has a specific native conformation (see below) that is characterized by β and γ dihedrals in trans and gauche+ (g+) orientations, respectively (Figure 3). The ffs rather prefer G_{S+1} phosphate in an alternative, i.e., flipped state, characterized by β_{g+} and γ_{trans} dihedrals. However, counterintuitively, disruptions of the UUCG TL are initiated from the native state of G_{S+1} phosphate and not from the alternative (likely spurious) flipped conformation. We suggest that the occurrence of the alternative backbone state is a consequence of structural strain in the native structure (in the ff description), which is reduced either by adopting the flipped backbone conformation or by onset of the overall TL structure disruption.

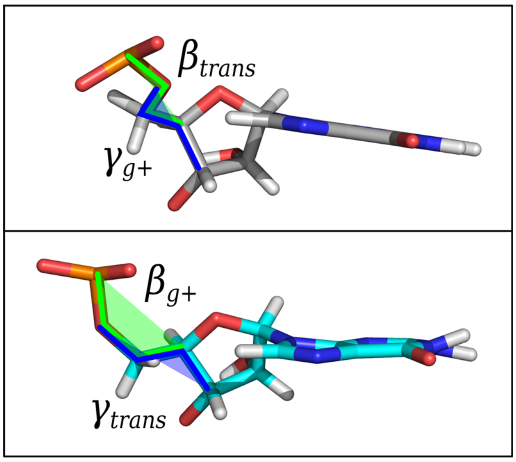

Figure 3. G_{S+1} phosphate in its native topology characterized by β and γ dihedrals in trans and gauche+ (g+) orientations (top) and in the flipped state characterized by β and γ dihedrals in g+ and trans orientations (bottom).

The other structural factors are the underestimated stability and subsequent loss of both signature sugar–base H-bonds within the loop, i.e., U_{L1}(2′-OH)···G_{L4}(O6) and U_{L2}(2′-OH)···G_{L4}(N7) H-bonds. The 2′-OH groups in both H-bonds form alternative transient H-bonds during standard MD simulations. The former H-bond is reversibly lost due to the flip of U_{L1}(2′-OH) toward the U_{L2} phosphate,[21] so that a transient U_{L1}(2′-OH)···U_{L2}(O5′) H-bond is established (Figure 2B). The U_{L2}(2′-OH)···G_{L4}(N7) H-bond is also reversibly lost, and the U_{L2}(2′-OH) is then either pointing toward the solvent or alternative U_{L2}(2′-OH)···G_{L4}(O6) H-bond is formed. Importantly, formation of the alternative U_{L2}(2′-OH)···G_{L4}(O6) H-bond is not in disagreement with NMR spectra (see below).[25]

Loss of the U_{L1}(2′-OH)···G_{L4}(O6) H-bond is accompanied with structural rearrangement of the G_{L4} nucleobase. G_{L4} nucleotide (in its *syn* state) forms the GU wobble base pair with U_{L1}. Once the U_{L1}(2′-OH)···G_{L4}(O6) H-bond is lost, G_{L4} makes a partial twist within the loop, where the signature base–base G_{L4}(N1H)···U_{L1}(O2) H-bond is weakened and a bifurcated G_{L4}(N1H/N2H)···U_{L1}(O2) H-bond is established instead (Figure 2B). This transient state is frequently observed in simulations, although the signature U_{L1}(2′-OH)···G_{L4}(O6) H-bond is typically reformed, followed by return of the G_{L4} nucleobase back to its canonical conformation. However, when the second sugar–base H-bond is lost simultaneously, i.e., neither the signature U_{L2}(2′-OH)···G_{L4}(N7) H-bond nor the alternative U_{L2}(2′-OH)···G_{L4}(O6) H-bond is formed, G_{L4} has an increased tendency to flip out of the loop (Figure 2C). Consequently, another structural feature, i.e., ribose–base stacking between the G_{L4} nucleotide and the O4′ atom of C_{L3} ribose, is lost. In other words, simultaneous loss of the signature interactions formed by both U_{L1}(2′-OH) and U_{L2}(2′-OH) groups opens the path for further destabilization of the structure, i.e., flip out of the G_{L4}. As mentioned above, the scenario when G_{L4} flips out of the loop requires G_{S+1} phosphate in its native state.

Once the G_{L4} is oriented toward the solvent (bulge-out state,[6,86] Figure 2D), it typically remains there, i.e., the disruption pathway appears to be irreversible within typical timescales of standard MD simulations (several microseconds). Nonetheless, recently published 10 μs long standard MD simulations with the χ_{OL3CP} + gHBfix19 ff version revealed occasional return of the G_{L4} nucleotide back from the bulge-out state to its native conformation within the loop.[19] We identified the same flip-out flip-in pathway also in simulations with the DESRES[31] RNA ff.[32] Thus, the results indicate existence of a rather narrow conformational corridor (bottle-neck) for expulsion and insertion of G_{L4}. It is worth noting that the tendency of expelling G_{L4} out of the loop is higher when another signature interaction within the loop, i.e., the C_{L3}(N4H)···U_{L1}(*pro*-R_P) base–phosphate (BPh) interaction type 7 (7BPh)[37] is weakened.

Conformation of the G_{L4}–G_{S+1} Sugar–Phosphate Moiety May Lead to Steric Clash of the 0BPh Interaction. As noted above, in simulations of folded UUCG TL, the G_{S+1} phosphate fluctuates frequently between two substates. The G_{S+1} phosphate flipping is mainly characterized by the shift of $\beta_{trans}/\gamma_{g+}$ dihedrals toward alternative $\beta_{g+}/\gamma_{trans}$ conformations (Figure 3). The flips can be monitored by the RNA suitename code[87] as they change backbone conformations (RNA suites) from the canonical **1a** to unusual **1e**, **1c**, or **unidentified** RNA suites. Before the loss of native UUCG TL topology, the G_{S+1} phosphate flipping is entirely reversible. In these initial parts of simulations, the alternative flipped state is prevalent (∼70% population, Figures S2 and S3 in the Supporting Information)[19] and is accompanied by loss of the G_{S+1}(C8H8)···G_{S+1}(O5′) 0BPh interaction (Figure 2A).

0BPh is a weak C–H···O H-bond established by the C6–H6/C8–H8 groups of the pyrimidine/purine nucleotides, which in most RNAs typically involves bridging O5′ phosphate oxygen from the same residue.[37] In **1a** suite conformation of canonical A-RNA duplexes, O5′ oxygens are often located below planes of the nucleobases. The unique structural context

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of UUCG TL characterized by the presence of neighboring G_{L4} nucleotide in syn conformation is causing a local structural change (deformation with respect to canonical **1a** suite of A-RNA duplexes) of sugar−phosphate backbone around the G_{S+1} nucleotide (Figure 4). According to the sugar−phosphate backbone nomenclature,[88] typical values of dihedrals around the G_{S+1} phosphate in its native UUCG conformation are on the edge of canonical **1a** suite definition (A-RNA-like) with the γ dihedral shifted toward higher values of ∼100°, similarly to the **1e** suite. The UUCG-specific conformation of G_{L4}−G_{S+1} sugar−phosphate moiety may be strained by potential steric clashes of G_{S+1}(O5′) oxygen with (i) G_{S+1}(C8−H8), i.e., the above-mentioned 0BPh interaction, and (ii) G_{S+1}(O4′) of ribose. The suspected 0BPh steric clash can be easily relaxed in the structural context of the A-RNA duplex by slight elongation of C8H8/C6H6···O5′ distances (Figure S4 in the Supporting Information). However, in the specific structural context of UUCG TL caused by the G_{L4} syn orientation, the suspected steric clash of the 0BPh interaction of the G_{S+1} phosphate can be relaxed only at the expense of intensifying the other steric clashes between G_{S+1}(O5′) and G_{S+1}(O4′) oxygens. As a consequence, the G_{S+1} phosphate is prone to flip toward the alternative (noncanonical) $\beta_{g+}/\gamma_{trans}$ state, where the 0BPh G_{S+1}(C8H8)···G_{S+1}(O5′) interaction is lost.

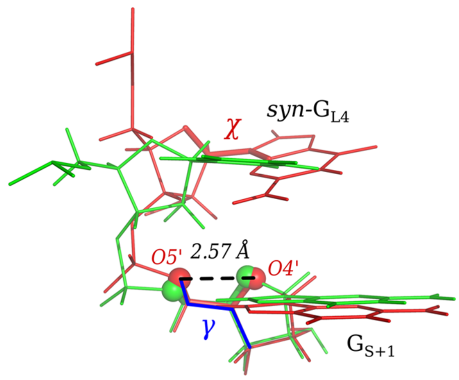

Figure 4. Local deformation of the sugar−phosphate backbone around the G_{S+1} phosphate caused by the syn conformation of the G_{L4} nucleotide (red) and comparison with the canonical **1a** suite of the A-RNA duplex (green). Both superimposed fragments are taken from the starting structure of the r(ggcacUUCGgugcc) 14-mer (PDB ID 2KOC[25]). The UUCG-specific conformation of the sugar−phosphate backbone (in red) contains two close contacts of the G_{S+1}(O5′) oxygen with (i) G_{S+1}(O4′) and (ii) G_{S+1}(C8H8) groups, which might cause steric clashes under the ff description and then lead to the G_{S+1} phosphate flip. The χ dihedral of the syn G_{L4} nucleotide, γ dihedral of the G_{S+1} nucleotide, and average distance in the NMR model[25] between O5′ and O4′ atoms of the G_{S+1} nucleotide (2.57 ± 0.014 Å) are highlighted. X-ray structures show slightly larger G_{S+1}(O5′)···G_{S+1}(O4′) distances with values of 2.64−2.71 Å.[38,89,90]

Unambiguous characterization of the G_{L4}−G_{S+1} sugar−phosphate moiety is difficult even for experiments. There are only a few nuclear Overhauser effect (NOE) intensities and no ^3J-scalar couplings available for detailed classification of this structural part by NMR spectroscopy.[25] We realized that some NOEs, namely, G_{S+1}(H8)−G_{S+1}(H4′) and G_{S+1}(H8)−G_{L4}(H3′), are violated even in the NMR structure. Other violations (considering H5′ and H5″ protons) can be explained by possible neglect of spin diffusion in the NMR experiment. In the MD ensemble, those parts of trajectories with native G_{S+1} phosphate conformation have better agreement with the experiment (for 10 NOE signals around the G_{L4}−G_{S+1} sugar−phosphate moiety) than those with flipped phosphate states except of two signals: (i) G_{L4}(H8)···C_{L3}(H5″) and (ii) G_{S+1}(H8)···G_{S+1}(H5″). X-ray structures show that RNA suite for the G_{L4}−G_{S+1} moiety is either **1a**,[38,89] **1c**,[90] **unidentified**,[38] or even **1L**.[89] Nevertheless, although the preference of RNA ff's for the alternative G_{S+1} phosphate conformation is likely an ff artifact, the experimental uncertainties around the G_{L4}−G_{S+1} moiety may indicate some real dynamics in this key part of the UUCG TL.

MM Description Leads to Excessive Short-Range Repulsion in the 0BPh Interaction. We analyzed the 0BPh interaction using electronic structure calculations. We compared QM/MM and MM behavior of two distances that characterize the 0BPh interaction, namely, the G_{S+1}(C8H8)···G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′) interactions. We took the UUCG 8-mer TLs, where the G_{S+1} phosphate maintained its native geometry (starting structures **6a−e**, **7a−e**, and **9a−e**, see the Methods section). Fifteen gradient geometry

Table 1. Comparison of the QM and MM Description of the G_{S+1}(C8H8)···G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′) Distances (in Å) within UUCG 8-mers and Dinucleoside Monophosphate Models[a]

	G_{S+1}(C8H)···G_{S+1}(O5′)			G_{S+1}(O4′)−G_{S+1}(O5′)		
UUCG TL	initial value[b]	$\Delta d_{QM/MM}$[c]	Δd_{MM}[d]	initial value	$\Delta d_{QM/MM}$[c]	Δd_{MM}[d]
structure **6e**	2.18	−0.09	0.10	2.74	0.04	0.09
structure **7e**	2.39	0.00	0.09	2.70	0.05	0.13
structure **9e**	2.15	0.06	0.15	2.57	0.05	0.11
dinucleoside monophosphate						
	initial value[b]	$\Delta d_{QM/COSMO}$[e]	$\Delta d_{MM/GB}$[f]	initial value	$\Delta d_{QM/COSMO}$[e]	$\Delta d_{MM/GB}$[f]
planar-native	2.37	−0.14	0.13	2.76	−0.05	0.11
tilted-native	2.18	0.04	0.26	2.74	0.00	0.06

[a]See the Methods section for the description of structures. UUCG TLs (structures **6**, **7**, and **9**) and dinucleoside monophosphate models (planar-native, tilted-native) possessing the native conformation of G_{S+1} phosphate were chosen for the analysis. UUCG TLs were optimized by QM/MM or MM, while the dinucleoside monophosphate models were optimized with geometry restraints by QM/COSMO or MM/GB. For UUCG 8-mer, values are shown only for structures **6e**, **7e**, and **9e**. See the Supporting Information for explanation and complete data, i.e., for 8-mer structures with different water distributions (**a−e**) and for dinucleoside monophosphate models optimized without geometry restraints (Section S4 and Tables S1−S3). [b]Initial values from the NMR model are 2.15 and 2.57 Å for G_{S+1}(C8H8)···G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′), respectively (structure **9**).[25] [c]$\Delta d_{QM/MM}$ = QM/MM value − initial value, i.e., change of the distance upon optimization. [d]Δd_{MM} = MM value − initial value. [e]$\Delta d_{QM/COSMO}$ = QM/COSMO value − initial value. [f]$\Delta d_{MM/GB}$ = MM/GB value − initial value.

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optimizations carried out from these starting structures showed that the RNA ff prefers longer distances in comparison with QM by ∼0.13 and ∼0.05 Å for the G_{S+1}(C8H8)···G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′) interactions, respectively (Tables 1, S3, and S4). Additionally, we used the dinucleoside monophosphate model (Figure S5 in the Supporting Information) possessing the native G_{S+1} phosphate conformation and analyzed its QM- and MM-optimized geometries. Again, both the G_{S+1}(C8H8)···G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′) distances are longer (by ∼0.26 and ∼0.10 Å) in MM compared to those in the QM description (Tables 1 and S5 in the Supporting Information).

We then performed MM interaction and conformational energy scans of the G_{S+1}(C8H8)···G_{S+1}(O5′) distance using two models: (i) a smaller dimethyl-phosphate−methyl-guanine molecular complex lacking the G_{S+1}(O4′)−G_{S+1}(O5′) contact and (ii) a larger O3′-methylated-guanosine-monophosphate intramolecular model, which included the G_{S+1} ribose with its O4′ oxygen atom and thus took into account the repulsion between G_{S+1}(O4′) and G_{S+1}(O5′) atoms (Figure S6A,B in the Supporting Information). The smaller model showed prolonged MM G_{S+1}(C8H8)···G_{S+1}(O5′) distance (by 0.11 Å) in comparison with QM (Figure 5). The difference doubles to

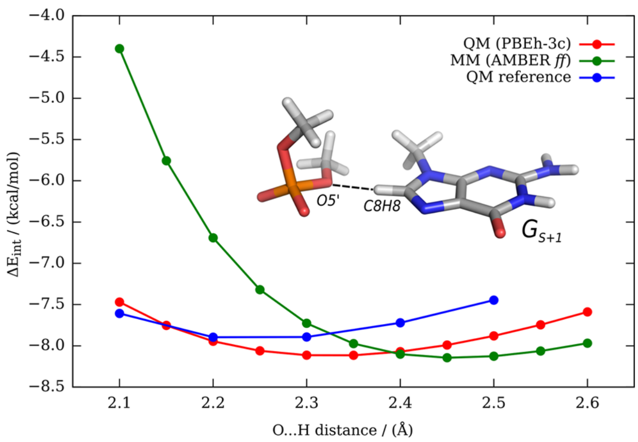

Figure 5. Comparison of QM (PBEh-3c; red), MM (AMBER ff—see the Methods section; green), and QM reference (DLPNO-CCSD(T)/CBS; blue) interaction energy profiles along the G_{S+1}(C8H8)··· G_{S+1}(O5′) H···O distance of dimethyl-phosphate−methyl-guanine model. The interaction energy scan was initiated from the G_{S+1} conformation found in the UUCG TL. Optimal distances of 2.32, 2.46, and 2.24 Å for QM, MM, and QM reference, respectively, show overestimation of the optimal G_{S+1}(C8H8)···G_{S+1}(O5′) distance by MM. Note that the MM description not only overestimates the optimum H-bond distance but also sharply exaggerates steepness of the short-range repulsion, which is a hallmark artifact of 6−12 Lennard−Jones potentials when describing close interatomic contacts and steric clashes.[91] Since the computations are done in the gas phase and the system is non-neutral, the magnitude of overall attraction is overestimated for all three methods compared to the same interaction in its complete environment. However, relative positions of the curves are expected to be unaffected by this simplification.

0.22 Å for the larger model (Figure S7 in the Supporting Information). We suggest that the increased difference (by 0.11 Å) obtained for the larger model is due to the G_{S+1}(O4′)−G_{S+1}(O5′) repulsion. In addition, the small-model calculations reveal that the PBEh-3c QM method used for the large-scale QM/MM calculations overestimates the G_{S+1}(C8H8)···G_{S+1}(O5′) H-bond distance by 0.08 Å with respect to the QM reference (Figure 5). In other words, the identified overestimation of short-range repulsion by the MM is even larger than indicated by the above comparison between PBEh-3c QM/MM and MM data.

In summary, the 0BPh interaction approximated by AMBER RNA ff is not consistent with QM. Both G_{S+1}(C8H8)··· G_{S+1}(O5′) and G_{S+1}(O4′)−G_{S+1}(O5′) contacts contribute to the imbalance. The optimal interatomic distance and mainly short-range repulsion (Figure 5) are severely overestimated. We propose that there is enough structural flexibility to overcome this inaccuracy for canonical A-RNA duplex conformation characterized by the **1a** suite. However, the discrepancy becomes unmasked in the restricted conformational space within UUCG TL formed by the G_{L4} syn nucleotide/G_{S+1} phosphate motif. Overestimated repulsion in MM approximation forces the G_{S+1} phosphate into flipped states seen during the MD simulations.

Relationship between the G_{S+1} Phosphate Conformation and Positioning of the G_{L4} Nucleotide. In MD simulations, the angle between planes of the G_{L4} syn base and of the closing $G_{S+1}C_{S-1}$ base pair fluctuates in the range of ∼10−30°. We call arrangements at the edges of the range as planar and tilted positions of G_{L4}. Both positions appear consistent with experimental data.[25,38,89,90] We carried out QM analyses of positioning of the G_{L4} syn base, which indicate that the native state of the G_{S+1} phosphate could excessively force G_{L4} nucleotide to the planar conformation during MD simulations. This conformation is prone to weakening of the signature U_{L2}(2′-OH)···G_{L4}(N7) sugar−base H-bond, which may contribute to problems in the MD description. However, we were not able to further quantify this potential MM imbalance due to the uncertainties inherent to the used computations. The full description is available in the Supporting Information.

Signature U_{L2}(2′-OH)···G_{L4}(N7) H-Bond Is Imperfectly Described by AMBER ff. The U_{L2}(2′-OH)···G_{L4}(N7) sugar−base H-bond (marked as "1" in Figure 1) is another player in the overall UUCG balance. According to structural data, it could alternate with the U_{L2}(2′-OH)···G_{L4}(O6) H-bond.[25] Melting experiments with various substitutions at the U_{L2}(2′-OH) position (preventing the H-bonding) suggested its only negligible effect on the thermodynamic stability of the loop.[92] Analysis of 10 μs long standard MD simulations[19] predicted that the U_{L2}(2′-OH)···G_{L4}(N7) H-bond is not very stable since the U_{L2}(2′-OH) is pointing either toward the solvent or is too far from both G_{L4}(N7) and G_{L4}(O6) acceptors in ∼25% of all snapshots with the native TL arrangement. When U_{L2}(2′-OH) is interacting with G_{L4} nucleobase, the native G_{L4}(N7) acceptor is preferred over G_{L4}(O6) (∼75 and ∼25%, respectively). In addition, even when formed, the U_{L2}(2′-OH)···G_{L4}(N7) H-bond shows poor directionality, with the O−H group pointing above the G aromatic moiety rather than directly to the N7 atom. This may be sign of a local imbalance of the ff description and understabilization of the U_{L2}(2′-OH)···G_{L4}(N7) interaction. Importantly, loss of the signature U_{L2}(2′-OH)···G_{L4}(N7) H-bond allows reconformation of the G_{L4} nucleobase within the loop and formation of the bifurcated G_{L4}(N1H/N2H)···U_{L1}(O2) H-bond. It is worth mentioning that the bifurcated G_{L4}(N1H/N2H)···U_{L1}(O2) H-bond is in agreement with some X-ray structures;[21,38,89] however, in simulations, its formation is typically followed by flip of the G_{L4} nucleobase out of the loop and subsequent disruption of the

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native state (Figure 2B). U_{L2}(2′-OH)⋯G_{L4}(N7) dynamics is thus directly kinetically related to the loss of the native state.

We employed a set of QM calculations to investigate this signature sugar−base H-bond. Initially, we compared donor−acceptor distances and directionality of the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond in QM/MM- and MM-optimized UUCG 8-mers (Tables S7 and S8 in the Supporting Information). We observed that although the QM and MM donor−acceptor distances are similar (Table S7), the O−H⋯N angle is smaller (by ∼18° in average) in MM compared to QM (Tables 2 and S8; Figure S9). Thus, we designed a small model (composed of the G_{L4} base and U_{L2} ribose; Figure S6C,D in the Supporting Information) and performed interaction energy scans of the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond. We used two different values of the O−H⋯N angle, i.e., (i) ∼170° (almost ideal angle for H-bond) and (ii) ∼143° (mimicking the H-bond often seen during MD simulations). MM minimum of interaction energy profiles is slightly shifted to shorter donor−acceptor distances (by ∼0.15 Å) and is slightly narrower in comparison with the broad QM minimum (Figure 6). Thus, any enforced prolongation of the O⋯N distance by, e.g., structural dynamics of other moieties within the loop would result in excessive weakening of the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond under MM description. In addition, the ff description obviously neglects the stabilizing effect of X−H bond stretching; this point is not analyzed by our calculations since the QM scan was done with rigid monomers, but it has been discussed elsewhere.[34,60]

Table 2. Description of Directionality of the Signature U_{L2}(2′-OH)⋯G_{L4}(N7) H-Bond (O−H⋯N Angle in deg) in QM/MM and MM Optimizations of the Whole TL 8-mers[a]

structure	initial angle	$\Delta angle_{QM/MM}$[b]	$\Delta angle_{MM}$[c]
1	143.4	16.1	5.2
2	118.1	45.2	20.8
4	140.8	14.4	−12.6
5e	161.7	9.3	2.5
6e	157.0	5.5	−7.0
9e	143.5	25.6	−0.9

[a]Structure **3** was excluded because the U_{L2}(2′-OH) forms an alternative H-bond with G_{L4}(O6) instead of the G_{L4}(N7) atom. Structures **7** and **8** were excluded because they contain spurious state of the U_{L1}(2′-OH) group, which may influence the position of the G_{L4} nucleobase and subsequently the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond. Values are shown only for structures **5e**, **6e**, and **9e** (see Table S8 in the Supporting Information for complete data, i.e., for 8-mer structures with different water distributions **a**−**e**). [b]$\Delta angle_{QM/MM} = angle_{QM/MM} - angle_{initial}$, i.e., change of the angle upon QM/MM optimization. [c]$\Delta angle_{MM} = angle_{MM} - angle_{initial}$.

More importantly, comparison of QM and MM PESs revealed that the effect of loss of directionality of this H-bond, i.e., comparing the almost ideal H-bond angle (∼170°) with the lowered one (∼143°), is underestimated (by ∼0.7 kcal/mol) by the MM method (Figure 6). Although MM still prefers (in the isolated small system) a H-bond with good directionality, the penalty for making the H-bond less directional is lower at the MM level compared to the QM level. Therefore, if some other energy terms in the UUCG TL profit from nonplanarity of this H-bond due to overall conflict among some other energy terms, the MM description may be excessively submissive to such strains.

Next, we used the U_{L2} ribose model and analyzed behavior of the C2′−O2′−O2′H angle and C1′−C2′−O2′−O2′H dihedral by QM and MM optimizations (Figure S10 in the Supporting Information). We performed two conformation energy scans: (i) C2′−O2′−O2′H angle bending and (ii) rotation of the C1′−C2′−O2′−O2′H dihedral. Both QM and MM conformational scans were comparable for bending of the C2′−O2′−O2′H angle (Figure S11 in the Supporting Information) and free optimizations showed a slightly shifted minimum (by 2.8°) between the QM and MM levels of theory (Table S9 in the Supporting Information). In contrast, conformational energy scans of the C1′−C2′−O2′−O2′H dihedral revealed differences especially around the region of interest, where the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond could be established (conformations with dihedral values between ∼150 and ∼210°, Figure 7). We identified a very flat QM minimum (∼190°) that is completely absent on the MM potential energy surface (PES). Instead, the MM curve is steeply descending in this region until it finds a deeper minimum at around 260° (Figure 7 and Table S9 in the Supporting Information). This result is in agreement with an analysis published by Mládek and co-workers[93] using larger sugar-to-sugar models (see Section S5 in the Supporting Information for more details).

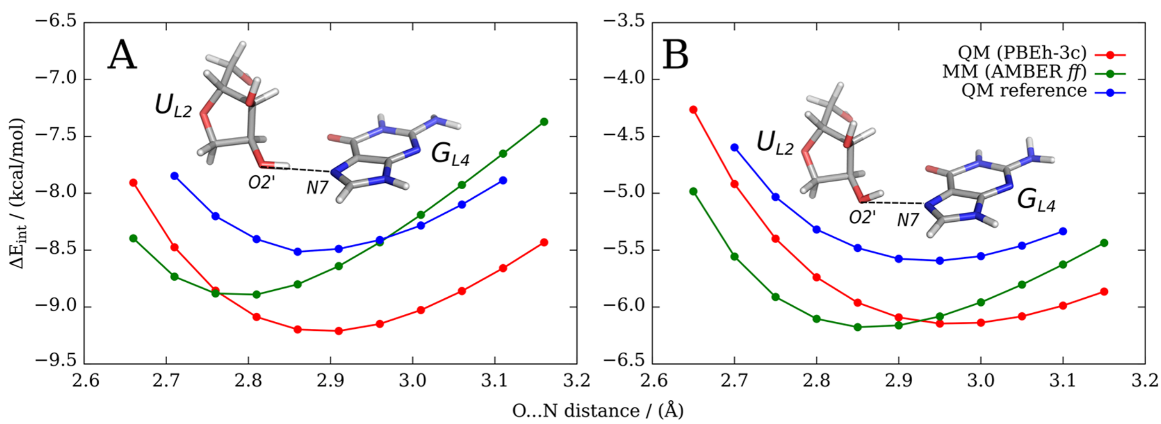

Figure 6. Interaction energy scans of the U_{L2}(2′-OH)⋯G_{L4}(N7) H-bond using QM (PBEh-3c; red), MM (AMBER ff; green), and QM reference (DLPNO-CCSD(T)/CBS; blue) methods. Ribose is shifted along the O⋯N vector in a distance range from 2.65 to 3.15 Å with a step length of 0.05 Å. All remaining degrees of freedom were frozen (see the Supporting Information for details). (A) O−H⋯N angle is 170° at the start of the scan. (B) O−H⋯N angle is initially 143°.

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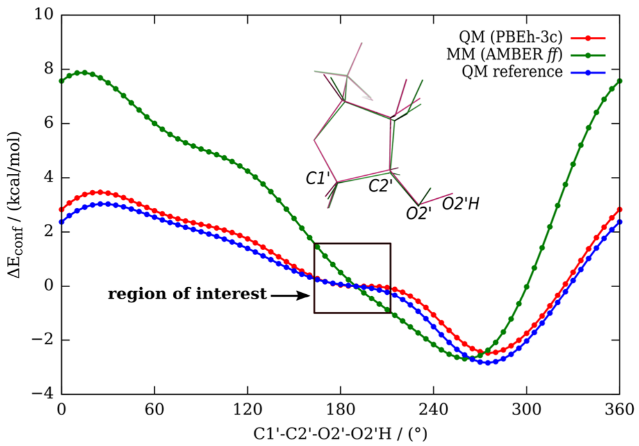

Figure 7. Conformation energy profile of the C2′-*endo* U_{L2} ribose C1′−C2′−O2′−O2′H dihedral using QM (PBEh-3c; red), MM (AMBER ff; green), and QM reference (DLPNO-CCSD(T)/CBS; blue) methods. Zero energy is set for a dihedral value of 190° (Figure S10 in the Supporting Information). The region of interest where the U_{L2}(2′-OH)···G_{L4}(N7) H-bond could be established is highlighted by the black rectangle. While QM finds a flat region around ∼190°, MM continues steeply to the global minimum at ∼260°, where the intramolecular O2′H···O3′ interaction is formed (Table S9 in the Supporting Information). See Section S4 in the Supporting Information for computational details and Section S5 and ref 93 for comparison with earlier scans of the 2′-OH group.

QM/MM and MM optimizations of UUCG TL 8-mers also revealed that the C1′−C2′−O2′−O2′H dihedral is larger in MM-optimized geometries (Table S10 in the Supporting Information), which is possibly associated with nonplanarity of the U_{L2}(2′-OH)···G_{L4}(N7) H-bond in MD simulations. Thus, all calculations indicate that both directionality and strength of the U_{L2}(2′-OH)···G_{L4}(N7) H-bond are not optimally described by the ff, which may contribute to common loss of this H-bond during MD simulations of UUCG TL. After the U_{L2}(2′-OH)···G_{L4}(N7) or alternative U_{L2}(2′-OH)···G_{L4}(O6) H-bonds are lost in MD simulations, the U_{L2}(2′-OH) group binds to another acceptor within the loop. Thus, it is meaningless to analyze the shift of the C1′−C2′−O2′−O2′H dihedral for such structures.

Any significant imbalance in description of the 2′-OH···N interactions could have major impact on RNA simulations, as these sugar−base H-bonds are abundant.[94] Further, correction of such imbalance would be difficult due to the lack of specific ff terms to affect the directionality of H-bonds. Thus, we performed a comparable analysis for the two 2′-OH···N1(A) H-bonds in an RNA kink-turn. However, in contrast to the UUCG TL case, these H-bonds behave stably (see Section S6, Table S11, and Figure S12 in the Supporting Information for details). We thus suggest that the accuracy of description of the 2′-OH···N interactions is generally acceptable for simulations of majority of RNAs and the moderate imbalance toward nonplanarity is unmasked for the UUCG TL due to its uniquely tight conformation.

U_{L1}(2′-OH)···G_{L4}(O6) Signature Sugar−Base H-Bond Is Weakened by Flips of the U_{L1}(2′-OH) toward U_{L2}(O5′). The signature U_{L1}(2′-OH)···G_{L4}(O6) sugar−base H-bond (marked as “2” in Figure 1) is crucial for the stability of UNCG TLs. It should be firmly established according to structural data[25,38,89,90] and any substitution at the U_{L1}(2′-

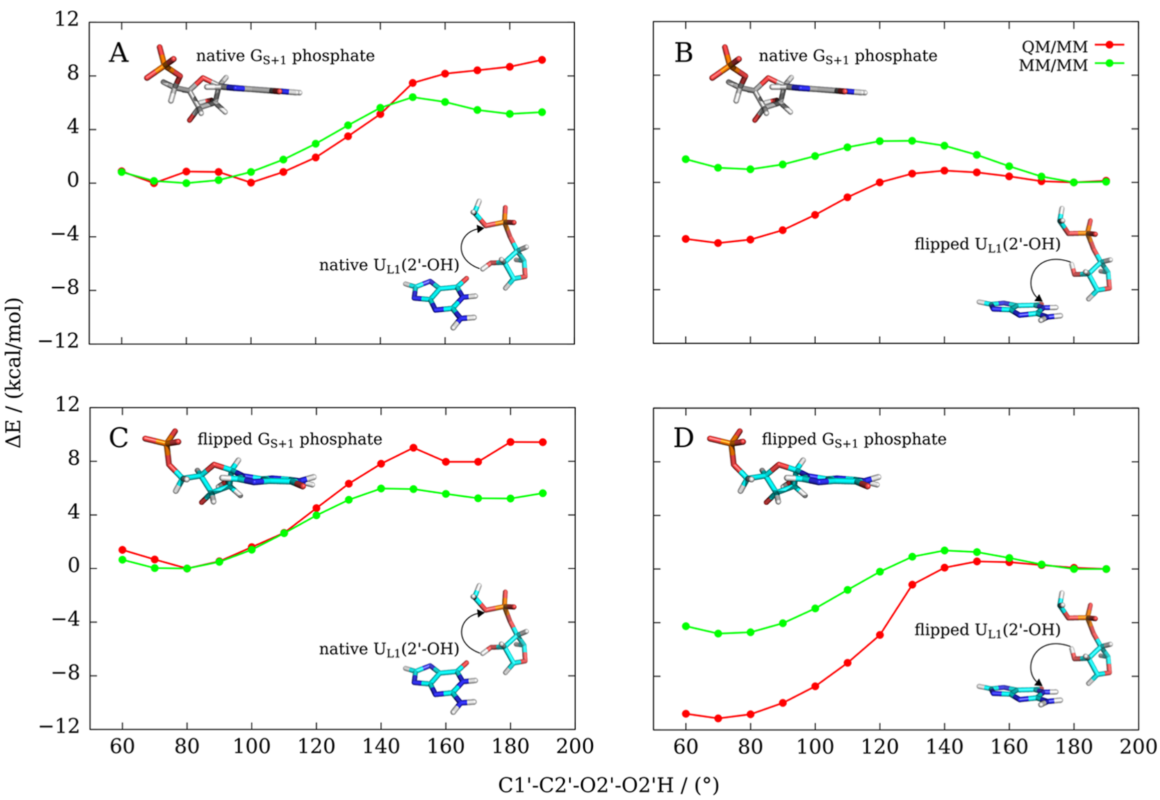

Figure 8. Comparison of conformational potential energy profiles of the U_{L1} ribose along the C1′−C2′−O2′−HO2′ dihedral for QM/MM (red) and MM/MM (green) methods. Initial structural orientations of the G_{S+1} phosphate and the U_{L1}(2′-OH) group are highlighted as structural insets, and directions of scans are shown as black arrows.

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OH) position leads to a significant drop of thermodynamic stability.[92] MD simulations show a reversible loss of the U_{L1}(2′-OH)···G_{L4}(O6) H-bond and transient formation of an alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond, a known ff artifact that was first classified in ref 21. We performed several standard MD simulations to understand this flip and its dependency on other factors. Although the signature U_{L1}(2′-OH)···G_{L4}(O6) H-bond is favored, there is a 10−30% population of the competing U_{L1}(2′-OH)···U_{L2}(O5′) H-bond. The latter H-bond is typically short-living (~hundreds of picoseconds), and the flips are entirely reversible. The flips were more frequent in the SPC/E solvent in comparison with TIP3P and OPC water models (Table S12 in the Supporting Information). Moreover, the flips were more common in 1.0 M KCl salt-excess ionic conditions than with 0.15 M KCl ionic strength and K^+ net-neutral ionic conditions. The modified vdW parameters for phosphate oxygens[45] did not bring any improvement (Table S12 in the Supporting Information).

We then investigated the competition between the signature U_{L1}(2′-OH)···G_{L4}(O6) and alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond using QM methods. First, we analyzed the QM/MM- and MM-optimized 8-mer TLs with native U_{L1}(2′-OH)···G_{L4}(O6) H-bond and observed comparable QM and MM description of this H-bond (structures **1−4**, **5a−e**, **6a−e** and **9a−e**). Then, we analyzed structures **7a−e** and **8a−e**, where the U_{L1}(2′-OH) group is flipped toward U_{L2} phosphate. For structure **7** having the native state of G_{S+1} phosphate, we observed renewal of the signature U_{L1}(2′-OH)···G_{L4}(O6) H-bond in all five (**a−e**) QM/MM optimizations, whereas in equivalent MM optimizations, the U_{L1}(2′-OH) group remained flipped toward U_{L2}(O5′) (see the Supporting Information for starting and final structures). Note that starting structures **a−e** used for optimizations differ only in water equilibration, while the RNA configuration is identical (see the Methods section).

Subsequently, we performed relaxed MM/MM and QM/MM scans (see the Methods section for details) of the UUCG TL 8-mers along the C1′−C2′−O2′−O2′H dihedral to quantify potential energy curves between the two conformations (Figure 8). We used MD snapshots with different states of the G_{S+1} phosphate and different orientations of the U_{L1}(2′-OH) group as starting points, namely, (i) the native state of the G_{S+1} phosphate and native orientation of the U_{L1}(2′-OH) group (structure **9**), (ii) the native G_{S+1} phosphate and flipped U_{L1}(2′-OH) group (structure **7**), (iii) the flipped state of the G_{S+1} phosphate and native orientation of the U_{L1}(2′-OH) group (structure **2**), and (iv) the flipped G_{S+1} phosphate and the flipped U_{L1}(2′-OH) group (structure **8**).

For all four scans (Figure 8), MM/MM calculations (green curves) overstabilize the structure with an alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond (the C1′−C2′−O2′−O2′H dihedral ~180°) with respect to the native U_{L1}(2′-OH)···G_{L4}(O6) H-bond (the C1′−C2′−O2′−O2′H dihedral ~80°), relatively to QM/MM data (red curves).

For UUCG TLs with native G_{S+1} phosphate conformation, we observed that going from the native H-bond (~80°) toward the alternative H-bond (~180°), the QM approach did not find any second minimum for ~180° (Figure 8A). In contrast, the alternative orientation of the U_{L1}(2′-OH) group toward U_{L1}(O5′) was recognized as a local minimum by MM (Figure 8A). Starting from the MD snapshot with the alternative H-bond, both QM/MM and MM/MM approaches identified the alternative state as a minimum. However, while for QM/MM scan, the native orientation of the U_{L1}(2′-OH) group was global minimum, the alternative orientation was global minimum of the MM/MM scan (Figure 8B).

Energy scans with the G_{S+1} phosphate in the flipped conformation (shift of β/γ dihedrals and absence of the G_{S+1}(C8H8)···G_{S+1}(O5′) 0BPh interaction) revealed the following picture. Starting from the native orientation of the U_{L1}(2′-OH) group, both QM/MM and MM/MM approaches indicated that global minimum is located near ~80° and the alternative orientation has no local minimum (Figure 8C). However, the MM/MM curve clearly underestimates the energy difference. Starting from the alternative state of the U_{L1}(2′-OH) group, both QM/MM and MM/MM indicated the native orientation as global minimum of the scan. However, the MM/MM curve is again substantially shifted in favor of the U_{L1}(2′-OH)···U_{L2}(O5′) H-bond relatively to the QM/MM curve (Figure 8D).

The MM/MM energy scans were performed with modified vdW parameters for phosphate oxygens (CP). We thus performed the MM/MM energy scan for structure **8** also without the CP modification. The difference was negligible (Figure S13 in the Supporting Information).

In summary, all PES scans showed that the alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond is overstabilized by the MM potential compared to QM/MM. In addition, in simulations, the spurious reversible flipping of the U_{L1}(2′-OH) group (~180°) is connected with dynamics of the G_{L4}-*syn* nucleotide by preferring the planar G_{L4} conformation. G_{L4} has, subsequently, a higher tendency to depart from the loop and collapsing it (see the paragraph above about disruption of the UUCG TL).

We note that energy scans should always be considered within the limits of sampling inherent to mere structure optimizations.[34] A detailed inspection of Figure 8 shows that there is an apparent inconsistency of (especially) the MM curves between panels A−B and C−D, i.e., between scans with the same G_{S+1} phosphate conformation. This indicates that other (still unidentified) structural factors are probably involved. We explicitly inspected those mentioned in our earlier study,[21] i.e., $\varepsilon(U_{L1})$, $\varepsilon(C_{S-1})$, $\zeta(C_{S-1})$, and $\zeta(U_{L1})$ dihedrals as well as sugar pucker of U_{L1}, but we did not find any clear structural changes along the PES curves. Thus, it appears that the flipped state of the U_{L1}(2′-OH) group might cause some structural adjustments distributed across many parameters that are difficult to identify (see the Supporting Information for initial and final structures from PES scans for comparison).

C_{L3}(O4′)···G_{L4}(Base) Ribose−Base Stacking Is Over-Repulsive in AMBER ff Relatively to QM. The CpG conformation in UUCG TL is identical to that found in Z-DNA and belongs to a "Z-like" motif. The characteristic feature of the "Z-like" motif is the perpendicular ribose−base stacking, where, for UUCG TL, the C_{L3}(O4′) points to G_{L4} nucleobase.[95] This contact represents a moderately attractive vdW interaction,[96−98] which, however, appears to be rather imposed by the overall structural context than being formed on its own.[98] As it includes an exceptionally short experimental distance between the O4′ atom and the nucleobase ring (~2.9 Å),[95] its MM description may be challenging, which may lead to destabilization of the position of the C_{L3} ribose above the G_{L4} nucleobase in MD simulations. Indeed, the loss of the ribose−base stacking during MD simulations is associated with the flip of the G_{L4} nucleobase out of the loop. Thus, we

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analyzed the distance of the C_{L3}(O4′) atom to the G_{L4} *syn* base plane in our QM/MM- and MM-optimized UUCG 8-mers. MM overestimates the O4′−guanine distance by ∼0.04 Å compared to QM/MM (Table S13 in the Supporting Information). Additional calculations using a small ribose−guanine stacking model (Figure S6E in the Supporting Information) showed that MM prefers a larger optimal distance (by ∼0.13 Å; Figure S15) compared to QM (PBEh-3c). In summary, short-range repulsion of the ribose−base stacking is overestimated by MM, which likely also contributes to disruption of the UUCG TL during MD simulations, especially when assuming that the ribose−base stacking is a consequence of the overall complex topology of UUCG TL.

Large-Scale Implicit Solvent QM Calculations Are Consistent with QM/MM Data. In addition to QM/MM and MM calculations performed in explicit solvent, we also tested the effect of implicit solvation for QM and MM geometry optimizations of the UUCG TL. We used COSMO implicit solvent for QM calculations that are compared with equivalent MM/GB calculations. The results obtained with implicit solvation are consistent with the above QM/MM and MM data in explicit solvent and are discussed in Section S7 of the Supporting Information.

Attempts to Improve the UUCG TL Simulations. We made several attempts to improve the UUCG simulations by simple corrections based on the above investigations. We tested REST2 folding simulations of the r(gcUUCGgc) 8-mer and standard simulations of the r(ggcacUUCGgugcc) 14-mer. The former test includes complete folding landscape, but the simulations could be affected also by stability of the two-base-pair stem, which may be underestimated by the ff. For the 14-mer, we monitored transitions between the fully native and G_{L4} bulged-out conformations.[6,32,86]

NBfix Correction of the 0BPh Interaction Maintains the G_{S+1} Native Phosphate Conformation. We first attempted to eliminate the G_{S+1} phosphate flipping. We modified the pairwise vdW parameters via breakage of combination (mixing) rules using the so-called nonbonded fix (NBfix).[99] We changed the minimum-energy distance of the Lennard−Jones potential (i.e., $R_{i,j}$ parameter) for the −H8···O5′− pair (i.e., between H5 and OR atom types) to 2.8808 Å. The depth of the potential well ($e_{i,j}$ parameter) was kept at its default value of 0.0505 kcal/mol. In other words, we decreased the sum of vdW radii between H8 atoms of all purine bases and O5′ oxygens of phosphates by 0.25 Å.

The effect of NBfix correction was significant for both interaction energy scan of the dimethyl-phosphate−methyl-guanine model (see section MM Description Leads to Excessive Short-Range Repulsion in the 0BPh Interaction) and MD simulations of the r(ggcacUUCGgugcc) 14-mer. For the former case, we changed the G_{S+1} −H8···O5′− pair of a small model, and it resulted in a significant improvement of MM interaction energies with shift of the optimal distance toward the QM level description (Figure S16). For the latter, the NBfix correction was applied for all six purine bases of the r(ggcacUUCGgugcc) 14-mer. The G_{S+1} phosphate favored its native geometry compared to the flipped one (∼80/20%, Figure S17 in the Supporting Information). The NBfix correction shifts the G_{S+1}(C8H8)···G_{S+1}(O5′) distance histograms toward shorter values (Figure S18 in the Supporting Information). Thus, elimination of the clash stabilizes the 0BPh interaction and the native phosphate moiety position, as intended. However, it does not bring any overall help to the UUCG TL simulations. Disruption of the loop in standard simulations is even accelerated once the native G_{S+1} phosphate conformation is stabilized; in one of three simulations, the G_{L4} left its binding pocket just after ∼35 ns (Figure S17 in the Supporting Information). Hence, elimination of the G_{S+1} phosphate flipping, which is partially easing apparent conformational stress within the UUCG loop, is actually speeding up the loss of the TL structure initiated by the G_{L4}-*syn* nucleotide departure. In other words, the local flipped backbone conformation "sequesters" G_{L4} in the pocket away from the global unfolding pathway. This is a typical example of compensation of errors, in this case affecting the lifetime of the native state. Thus, despite the lack of overall improvement in simulations, we consider the modification of the 0BPh interaction as correct per se.

gHBfix alone Is Insufficient to Fold the gcUUCGgc TL. gHBfix potentials are general ff terms added to the basic ff for tuning stability of H-bonds. In our previous work, we have shown that combining the basic χ_{OL3CP} ff with the gHBfix19 improved RNA simulations without side effects, although it was still insufficient to fold the UUCG TL.[19]

We have thus tried several additional gHBfix settings to specifically improve folding of the r(gcUUCGgc) 8-mer. The most complex version strengthened base−base H-bonds and weakened sugar−phosphate interactions comparably to the gHBfix19 potential. In addition, it also strengthened the sugar donor−base acceptor H-bonds (as they represent key signature H-bonds in the UUCG TL) and weakened the base donor−sugar acceptor and sugar−sugar H-bonds. This version is abbreviated as $gHBfix_{UNCG19}$ (see the Supporting Information for details). Occasional UUCG folding events were observed with $gHBfix_{UNCG19}$, but overall population of the folded TL 8-mer remained negligible. Full methods and results are summarized in the Supporting Information.

Proposed Pathway of the G_{L4} Departure and Insertion. Although the $gHBfix_{UNCG19}$ variant is not sufficient to correctly fold the r(gcUUCGgc) 8-mer, it gives interesting insights. We analyzed all available REST2 simulations of the r(gcUUCGgc) 8-mer as well as the r(ggcacUUCGgugcc) 14-mer standard simulations showing some capability of G_{L4} to return back to the native position after its initial departure, i.e., from the bulge-out state[6,86] back to the native state.[19,32] The folding (and refolding) pathway requires a conformation where the G_{S+1} phosphate is in its native state. Thus, the identified ff imbalance between native and flipped conformations of the G_{S+1} phosphate is kinetically stabilizing the broader native basin (including the flipped G_{S+1} phosphate) once reached, as discussed in the previous section. However, it also hinders folding of the TL as the folding pathway goes through a bottleneck with the native G_{S+1}, which is enthalpically disfavored by the ff due to the steric clash associated with the 0BPh interaction (Figure 9). This explains why attempts to tune solely the stability of H-bonds were insufficient to correct the free-energy imbalance between native and misfolded states of the r(gcUUCGgc) TL.

Final Set of REST2 Simulations Shows Improvement in the Folding of the 8-mer UUCG TL, But It Is Still Not Sufficient. As an additional effort, we took the following ff version: standard χ_{OL3CP} RNA ff combined with the $gHBfix_{UNCG19}$ correction for H-bonds and NBfix correction for the −H8···O5′− atom-pair clash. In addition, we reduced the vdW radii of all nonpolar H atoms (H1, H4, H5, and HA atoms) universally to 1.2 Å. We have tried this additional vdW

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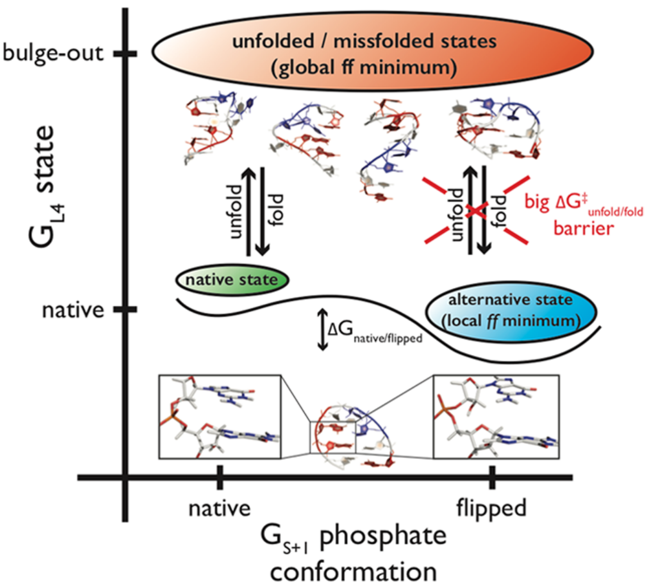

Figure 9. Proposed scheme describing folding and unfolding pathways of the r(gcUUCGgc) TL in MD simulations based on key conformational states of the G_{S+1} phosphate and G_{L4} nucleotide. The main conformational states are highlighted: (i) the native state with both G_{L4} and the G_{S+1} phosphate in the native conformation, (ii) the alternative state (local minimum in ff) with G_{L4} in the native and the G_{S+1} phosphate in the flipped conformation, and (iii) global ff minimum including all misfolded and unfolded states. Both folding and unfolding pathways require the G_{S+1} phosphate in its native conformation. The diagram explains why eliminating just the flipped G_{S+1} phosphate structures does not help to improve the simulations. The representative native structure of the r(gcUUCGgc) TL is shown at bottom, and both native and flipped G_{S+1} phosphate conformations are highlighted as insets. Some examples of misfolded structures often found in MD ensembles are also shown (top). G, C, and U nucleotides are in red, white, and blue, respectively.

modification as it seemed to us that these hydrogens may cause some steric conflicts in the native UUCG conformation, similarly to the 0BPh interaction. The depths of Lennard–Jones potentials were not modified. All of these fixes were aimed to stabilize the 0BPh interaction and further ease some possible—yet unidentified—steric clashes within the UUCG loop. We performed 20 μs long REST2 folding simulation of the r(gcUUCGgc) 8-mer (all replicas initiated from the completely unfolded single stranded structure). There was an increased probability of folding (Figure S20 in the Supporting Information), although the overall population of the native state within the reference replica at 298 K was still low. We obtained ~33% population of structures with correct stems and ~7% population of entirely correct native structure. This means that the population of the correctly structured loop in the subensemble with correct stem formation was ~20%. Thus, the introduced corrections increased the propensity of folding but were still not able to fully stabilize the native state within the reference replica. Nevertheless, it is one of the best folding simulations that has been achieved for the 8-mer UUCG TL system by any currently available ff.

As the last attempt, we carried out another REST2 simulation starting now from the native state (in all replicas), where we added to the above potential a restraint (see the Supporting Information for details) to the U_{L1}(C1′–C2′–O2′–O2′H) dihedral to prevent the flipping of the U_{L1}(2′-OH) group. It aimed to stabilize the signature U_{L1}(2′-OH)···G_{L4}(O6) H-bond instead of the alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond. In this simulation, the native state was lost in all replicas after ~7.5 μs (Figure S21 in the Supporting Information) with no tendency for refolding, which may indicate that the dihedral restraint caused some undesired side effects.

CONCLUDING REMARKS

The compact and well-determined native conformation of UNCG TL has served as a hallmark reminder of persisting problems of RNA ff's for structural description of RNA molecules.[4,6,8−11,19,21,30,86] In this work, we clarify why this RNA motif is so difficult for MD simulations.

We first inspected in detail the disruption (unfolding) paths of the UUCG motif during standard as well as enhanced sampling MD simulations and identified common rearrangements that precede disruption of the UUCG native topology. Collapse of the loop is characterized by a gradual loss of several key structural features: (i) G_{L4}–G_{S+1} phosphate moiety conformation, (ii) G_{S+1}(C8H8)···G_{S+1}(O5′) 0BPh interaction, (iii) C_{L3}–G_{L4} ribose–base stacking interaction, and (iv) both U_{L1}(2′-OH)···G_{L4}(O6) and U_{L2}(2′-OH)···G_{L4}(N7) sugar–base H-bonds.

We then performed numerous QM/MM and high-level QM calculations that reveal diverse limitations of the empirical ff potential. In particular, neither directionality nor strength of the signature U_{L2}(2′-OH)···G_{L4}(N7) H-bond is correctly described by the ff. The second signature U_{L1}(2′-OH)···G_{L4}(O6) H-bond is weakened by the formation of the alternative U_{L1}(2′-OH)···U_{L2}(O5′) H-bond, which is over-stabilized by the ff. Flipping of the U_{L1}(2′-OH) group (from base to phosphate) is connected with moderate relocation of the G_{L4}-*syn* nucleotide within the binding pocket. Subsequently, G_{L4} has higher tendency to depart from the loop. Further tests indicate that the ribose–base stacking interaction also suffers from a problematic MM description.

Collapse of the UUCG TL is connected with intrinsic dynamics of the G_{L4}–G_{S+1} sugar–phosphate moiety. Experimental characterization of this region is ambiguous as there are only few direct experimental observables[25] and some NOEs are violated even in the NMR-based models. MD simulations show that there are two states of G_{S+1} phosphate, i.e., native and flipped one, characterized by the shift of $\beta_{trans}/\gamma_{g+}$ dihedrals toward alternative $\beta_{g+}/\gamma_{trans}$ conformations. Thus, it appears that there is some intrinsic dynamics within this key part of UUCG TL, which could be an analogy to the recent indication of dynamic behavior of the G_{L4}-*syn* nucleotide.[86] Nevertheless, preference of RNA ff's for the alternative G_{S+1} phosphate conformation is clearly an ff artifact caused by incorrect description of the G_{S+1}(C8H8)···G_{S+1}(O5′) 0BPh interaction. Its optimal interatomic distance and short-range repulsion are severely overestimated in ff's, forcing the G_{S+1} phosphate into the flipped state seen during MD simulations.

Although being spurious, population of the flipped G_{S+1} substate kinetically decelerates unfolding. This is because folding (and refolding) pathways transit through a conformation having the G_{S+1} phosphate in its native state. Hence, also folding pathways reaching the (artificial) local minimum involving the flipped G_{S+1} phosphate have to pass through the native state conformation, which is enthalpically disfavored by the overestimated G_{S+1}(C8H8)···G_{S+1}(O5′) steric clash in AMBER RNA ff's. This is probably the reason why all our previous attempts of tuning solely the H-bond interactions

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were insufficient to correct the free-energy imbalance between native and misfolded states of the r(gcUUCGgc) TL.[19]

We attempted to fix the 0BPh clash by NBfix correction, to stabilize the native G_{S+1} phosphate conformation. This, however, accelerated disruption of the loop in standard simulations. Nevertheless, coupling this correction with the $gHBfix_{UNCG19}$ potential specifically designed to support H-bonds important for the UUCG TL improved REST2 folding simulations of the r(gcUUCGgc) 8-mer. We achieved (at 298 K) ~7% total population of the fully correct (both loop and stem) native structure, while the population of the correctly structured loop upon stem formation was ~20%. It is fair to admit that this result was still not fully satisfactory and was achieved by extended ff modifications that might cause side effects for other RNA systems, which we did not test.

In conclusion, UUCG TL remains a considerable challenge for atomistic simulations with classical ff's. We identified a number of individual energy imbalances. Our data could serve as an important guidance for tuning of pair-additive as well as polarizable ff's.[100,101] However, when trying to design ff corrections based on this knowledge, we still could not get a decisive improvement. There could be several reasons for this. For example, there could be further imbalances at the stem/loop interface that remain to be identified. Additionally, the extent of how simulations are affected by the uncorrected ribose−base stacking interaction is unknown. At last, a more fundamental problem could be the simplicity of the ff per se. Even when we correctly identify the imbalances, it could be challenging to suggest their correction with simple MM terms that fail to mimic the true electronic structure effects. As the key result, our work reveals that the poor behavior of UUCG simulations cannot be explained by a singular dominant factor that would be straightforwardly correctable. There is a concerted effect of multiple and mutually coupled ff inaccuracies, acting in the context of uniquely tight conformation of the UUCG TL, where the RNA chain does not provide a sufficient conformational flexibility to absorb or circumvent the individual imbalances. In addition, tuning the simulation performance of the UUCG TL is complicated due to enormous computational demands of the folding simulations and the difficulty to reach their quantitative convergence.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.0c00801.

Details on the MD protocol, list of small models, summary all QM and QM/MM calculations, additional computational details of small models, additional details about conformation scans of the 2′-OH group, computational results of the kink-turn Kt-7, details on large-scale QM calculations in the implicit solvent, relationship between the G_{S+1} phosphate conformation and positioning of the G_{L4} nucleotide, additional results from folding of the r(gcUUCGgc) TL using the gHBfix potential, description of the U_{L1}(C1′−C2′−O2′−O2′H) dihedral restraint, and supporting tables and figures (PDF)

Starting and final structures from QM/MM and MM optimizations (PDF)

Starting and final structures from PES scans (PDF)

Video showing the common UUCG disruption pathway during MD simulations (MP4)

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Notes

The authors declare no competing financial interest.

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